Shape and Space 5 — Transformations: translation, reflection, rotation, enlargement

Squared paper (and tracing paper for rotations) needed. Label every image. To *describe* a transformation, give all the details: vector / mirror line / angle, direction and centre / scale factor and centre. Marks in brackets. ★ = harder question.

1 Draw axes from -6 to 6. Draw triangle J with vertices $(-2, 3)$, $(2, 3)$ and $(1, 5)$. [8]

- **a** Translate J by the vector $\begin{pmatrix} 3 \\ -3 \end{pmatrix}$. Label the image K.
- **b** Reflect K in the line $y = -x$. Label the image L.
- **c** Reflect L in the line $x = -3$. Label the image M.
- **d** Describe fully the single transformation that maps M onto J.

a Add 3 to x, subtract 3 from y: K is $(1, 0)$, $(5, 0)$, $(4, 2)$.
b Reflection in $y = -x$ sends $(x, y) \to (-y, -x)$: L is $(0, -1)$, $(0, -5)$, $(-2, -4)$.
c Reflection in $x = -3$ sends $(x, y) \to (-6 - x,\ y)$: M is $(-6, -1)$, $(-6, -5)$, $(-4, -4)$.
d $M \to J$ is a **rotation of $90°$ anticlockwise about $(-6,\ 3)$**. Check with tracing paper, or: rotating $(-6, -1)$ by $90°$ anticlockwise about $(-6, 3)$ gives $(-2, 3)$ ✓ and $(-4, -4)$ gives $(1, 5)$ ✓.

2 Draw axes from -6 to 6. Draw triangle A with vertices $(-2, -1)$, $(-1, -1)$ and $(-1, 1)$. [12]

- **a** Rotate A $90°$ clockwise about $(1, 2)$. Label the image B.
- **b** Enlarge B by scale factor 2, centre $(-4, 6)$. Label the image C.
- **c** Translate C by $\begin{pmatrix} 2 \\ -6 \end{pmatrix}$. Label the image D.
- **d** Rotate D $90°$ anticlockwise about $(-3, -5)$. Label the image E.
- **e** Describe fully the single transformation that maps E onto A.

a B: $(-2, 5)$, $(-2, 4)$, $(0, 4)$.
b Double each point's distance from $(-4, 6)$: C: $(0, 4)$, $(0, 2)$, $(4, 2)$.
c D: $(2, -2)$, $(2, -4)$, $(6, -4)$.
d E: $(-6, 0)$, $(-4, 0)$, $(-4, 4)$.
e E has sides twice as long as A and the same orientation, so it's an **enlargement, scale factor $\frac{1}{2}$, centre $(2, -2)$**. (Lines joining corresponding vertices, e.g. $(-6, 0) \to (-2, -1)$ and $(-4, 4) \to (-1, 1)$, meet at $(2, -2)$.)

3 The point $P(a,\ b)$ is mapped to $Q(10,\ 10)$ by the following transformations *in this order*: reflection in $y = x$; rotation of $90°$ clockwise about O; translation by $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$; enlargement of scale factor 2 about O. Find a and b. [5]

> Track the point: reflection in $y = x$: (b, a). Rotation $90°$ clockwise about O sends $(x, y) \to (y, -x)$: $(a, -b)$. Translation: $(a - 2,\ -b + 4)$. Enlargement ×2: $(2a - 4,\ -2b + 8) = (10, 10)$. So $2a - 4 = 10 \Rightarrow a = 7$ and $-2b + 8 = 10 \Rightarrow b = -1$. $P(7, -1)$.

Total: 25 marks

Describe fully: translation → vector · reflection → mirror line equation · rotation → angle, direction, centre · enlargement → scale factor, centre. Useful rules: $y = x$: $(x, y) \to (y, x)$; $y = -x$: $(x, y) \to (-y, -x)$; $90°$ clockwise about O: $(x, y) \to (y, -x)$; $90°$ anticlockwise about O: $(x, y) \to (-y, x)$.